ADAPTIVE SMART GRID MANAGEMENT VIA GENTWIN & LLM INTEGRATION

A Transparent and Explainable Digital Twin Framework for Smart Grids

“Built upon the GenTwin Framework for Adaptive IoT Management”

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INTRODUCTION-GENTWIN

GenTwin: Motivation

Growing Problem in IoT Networks

The rapid expansion of IoT-based smart services has introduced:

- highly dynamic network topologies
- heterogeneous and distributed data sources
- frequent link/node failures
- strict real-time response requirements

These conditions make **adaptive network management** extremely difficult using traditional approaches.

Limitations of Current Solution

Limitation 1:

Traditional twins rely on historical data and fail on unseen scenarios.

Limitation 2:

Standard AI architectures are rigid; they require manual software updates for new data types.

Limitation 3:

High network dynamicity causes slow, inefficient adaptations in traditional models.

GenTwin Novelty (Proposed Solution Idea)

To address these issues, GenTwin introduces:

Priority Pooling

selects and ranks the most important IoT relations to handle limited LLM context efficiently

Twin Adapter

fine-tunes the LLM for digital twin modeling without full retraining

Enables **adaptive, scalable**, and **context-aware** IoT digital twin reasoning in real time

System Architecture Overview(contd)

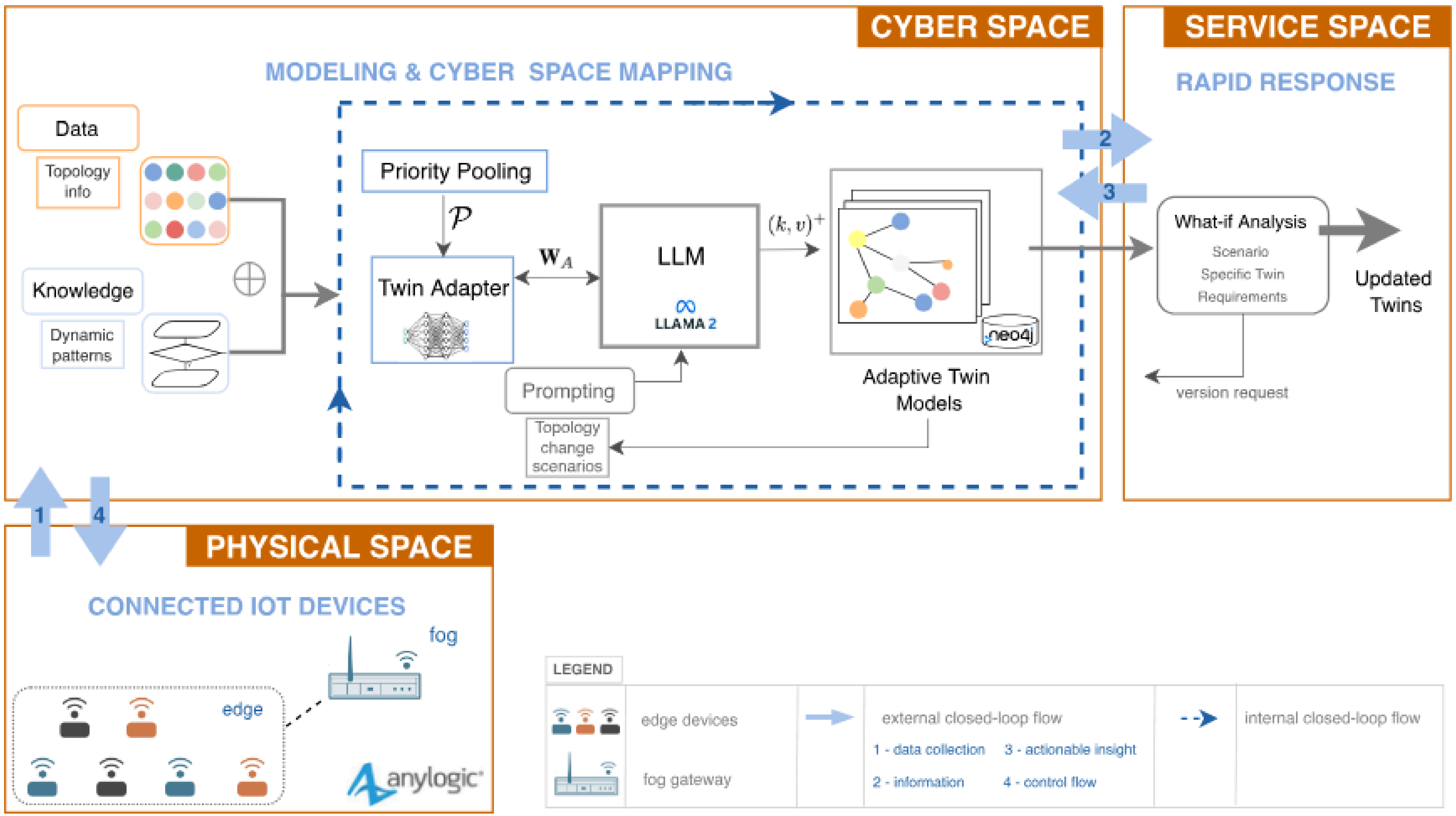


Fig. 1. Architectural framework of generative AI-aided digital twining - GenTwin.

GenTwin Paper's proposed system architecture

Motivation – The Hidden Problem

Modern smart grids are highly optimized systems that efficiently manage electricity demand and supply.
However, from the user's perspective, these systems behave like a black box.

- Users receive electricity bills without understanding the underlying reasons
- Devices may be automatically controlled without clear explanation
- Pricing changes dynamically, but users are not informed why

This lack of transparency creates a trust gap between the system and the consumer.

Problem Statement

Current smart grid systems lack an explainability layer that connects system decisions with human understanding.

Key challenges include:

- **Decision Opacity:** Optimization outputs are not interpretable
- **Context Overload:** Massive IoT data cannot be directly understood
- **No Natural Interface:** Users cannot query or understand grid behavior

Research Objective

This research proposes a GenTwin-based Explainable Smart Grid Framework that enables:

- Transparent understanding of grid decisions
- Natural language interaction between user and system
- Reconstruction of “**why**” behind system actions

Unlike traditional work, this system prioritizes:

Explainability over optimization

Core Idea (Conceptual Flow)

The proposed system introduces a new pipeline:

Physical Grid → GenTwin → LLM Reasoning → Explanation → User

- Real-world data is mirrored into a digital twin
- The twin simulates and interprets system behavior
- The LLM generates human-understandable explanations

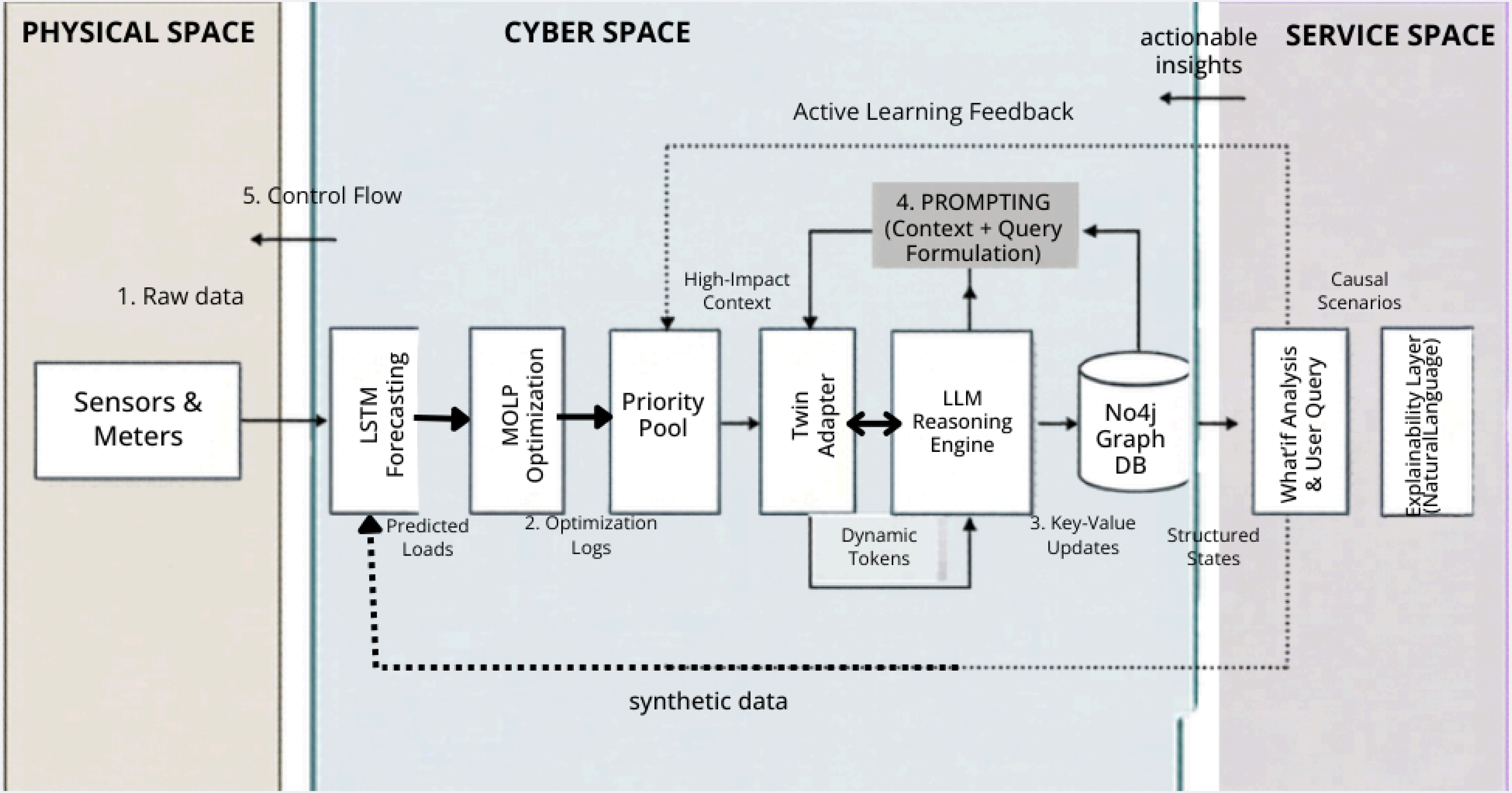
“We move from prediction to understanding.”

System Architecture Overview

The system is divided into three layers:

1. **Physical Layer**
   - Collects real-world data from smart meters, transformers, and weather sources
2. **Cyber Layer (GenTwin)**
   - Processes and interprets data
   - Builds a digital representation of the grid
   - peforms prediction and optimization
3. **Application Layer**
   - Provides explanations, and user interaction

Our System Architecture Overview



Physical Layer

The physical layer represents the **real-world smart grid environment**.

It collects:

- Energy consumption data from smart meters
- Environmental data from sensor such as temperature and humidity
- Grid-level data such as voltage, load, and pricing signals

However, this data is **raw** and **unstructured**, making it difficult for users to interpret directly.

Cyber Layer — GenTwin (Core Component)

The GenTwin acts as an intelligent digital replica of the physical grid.

It performs:

- State representation of the current grid condition
- Simulation of possible scenarios
- Reconstruction of past events

Unlike traditional digital twins, GenTwin integrates:

- LLM reasoning
- Domain knowledge via Twin Adapter
- Context filtering via Priority Pool

Role of LSTM (Supporting Component)

Input Data:

Historical load, weather conditions, calendar features, storage state from Knowledge Graph

Purpose:

Feed predicted demand into MOLP for real-time optimal allocation

Output:

Next-slot demand forecast

Training benefit:

What-if analysis generates synthetic rare-event data to improve forecast robustness

Priority Pool Mechanism

Due to the massive volume of IoT data, it is not feasible to process everything through the LLM.

Priority Pool mechanism:

- Ranks data based on criticality score
- Selects only relevant and high-impact events
- Filters noise and reduces context overload

This ensures efficient and meaningful reasoning by the LLM.

Role of LLM : Semantic Bridge

The LLM acts as a semantic bridge between system behavior and human understanding.

Its functions include:

- Translating user queries into system constraints
- Performing causal reasoning over grid events
- Generating natural language explanations

Example:

User Query:
"Why did electricity bill increase?"

System Response:
Likely AC usage probability = high

Twin Adapter & Knowledge Graph

The Twin Adapter converts smart-grid data into structured knowledge

01. Uses smart meter readings and grid topology

02. Stores real-time and historical grid state

03. Supports causal reasoning and explainability

04. Fine-tuned using domain-specific smart grid data

What-if Analysis as Active Learning & Decision Support

Generator Failure

Simulate sudden EP offline. Which consumers are affected? Which storage units can compensate? Reallocate immediately.

Storage Outage

Battery storage drops to zero. How does the grid cope during night hours with no solar? Which SGOs face shortage?

Demand Spike

Consumer demand surges 50% above forecast. MOLP re-optimizes allocation in real time using updated LSTM output.

Price Change

Real-time pricing jumps. Does consumer QoE change? Which allocation strategy minimizes cost impact?

Renewable Surge

Unexpected solar overproduction. Storage fills up. How to safely redirect excess energy without waste?

What-if Analysis does THREE things in GenTwin-SG:

① Identifies vulnerable grid nodes → increases their GCS weight

② Generates synthetic rare-event samples → improves LSTM training data

③ Tests MOLP decisions under stress → validates optimization quality

Why This Matters

This work shifts the focus of smart grid research:

From:

- System efficiency
- Cost optimization

To:

- User understanding
- Transparency
- Trust

A transparent system enables users to make informed decisions about their energy usage.

Challenges

Causal Explainability vs. Correlation

LLMs capture statistical correlations rather than true causal relationships, which may produce logically incorrect explanations.

Grounding LLM Explanations

LLM outputs must be strictly grounded in verified Neo4j knowledge graph data to prevent hallucinated explanations.

Prediction Error Propagation

Errors in upstream modules propagate through the sequential pipeline, affecting downstream optimization and explanation accuracy.

Knowledge Graph Completeness

Sensor failures and missing data can lead to incomplete or outdated knowledge graphs, reducing explanation reliability.

What-If Analysis Realism

Synthetic scenarios must comply with physical grid constraints to ensure realistic and reliable model training.

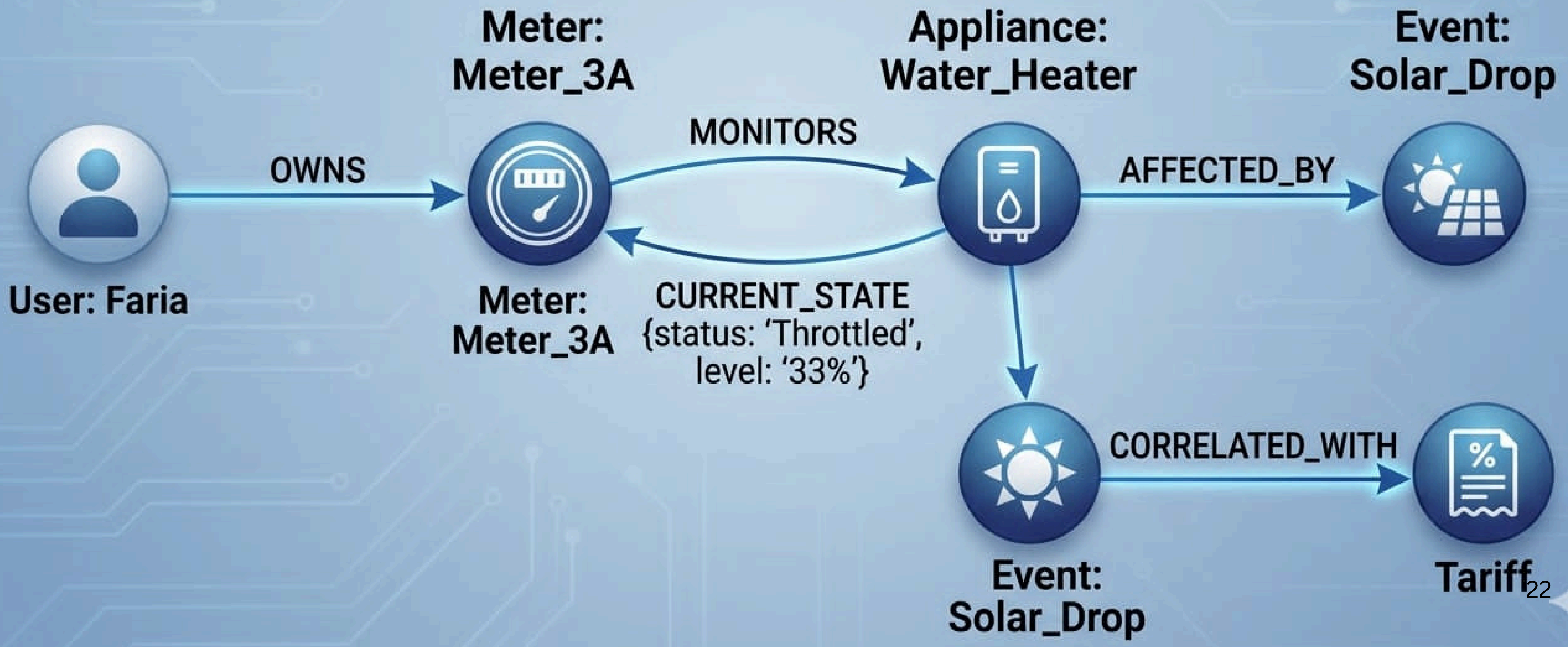
Dynamic Knowledge Graph Maintenance

Maintaining real-time graph consistency, concurrency control, and version management is a significant systems challenge.

THANK YOU

GRAPHICAL REPRESENTATION OF NEO4J KNOWLEDGE GRAPH

DETAILED VIEW OF SELECTED RELATIONSHIPS



Critical / Non-Deferrable (Oxygen Concentrator)

Node Centrality Weights

Curtailable / Thermostatically Controlled (Air Conditioner)

Schedulable

(User: Faria) —[OWNS]—► (Meter: Meter_3A)
(Meter: Meter_3A) —[CONTAINS]—► (Appliance: AC)
(Appliance: AC) —[HAS_STATE]—► (State: Throttled_50%)
(State: Throttled_50%) —[TRIGGERED_BY]—► (Event: Solar_Generation_Drop_30
(State: Throttled_50%) —[RESULTED_IN]—► (Benefit: Saved_15%_Cost)

Non-Intrusive Load Monitoring (NILM) :

Instead of requiring expensive sensors on every single home appliance, NILM analyzes the total voltage/ current waveforms from the main smart meter and mathematically splits it into appliance-specific tracking profiles (e.g., separating the base load from a curtailable AC or a non-deferrable oxygen concentrator).

Priority Pooling

Priority Pooling extracts:

- important entities,
- dynamic relations,
- topology relationships,
- interaction patterns.

Instead of storing only raw sensor values,
GenTwin learns:

- semantic relationships,
- entity dependencies,
- topology behavior.

So even if:

- some sensor data disappears,the system still understands:
- how entities are related.

Generative Reconstruction

smart meter of House-A disconnects temporarily.
But system still knows:

- historical usage pattern,
- nearby transformer load,
- weather condition,
- peak-hour pattern,
- neighboring household behavior.

Then GenTwin may estimate:
probable demand state of House-A.
Thus:

- KG remains partially complete,
- optimization continues,
- explanations still possible.

Why This Works Scientifically

GenTwin is NOT purely: “data storage.” It is: “relationship-aware adaptive modeling.”
That is why it survives fluctuating network conditions better than traditional twins.